

AIML SESSION 11 ASSIGNMENT SUBMISSION

Name of Student : Vaibhav Vijay Kate

Domain/Subject : AIML (Artificial Intelligence & Machine Learning)

College : Government Polytechnic College Washim

Branch : Information Technology

Session : 11

Q1

Q1. Basic try-except)

Write a program that takes two numbers as input from the user and prints their division.

Use try-except to handle the case when the user enters non-numeric input (ValueError).

Print a friendly message if an error occurs.

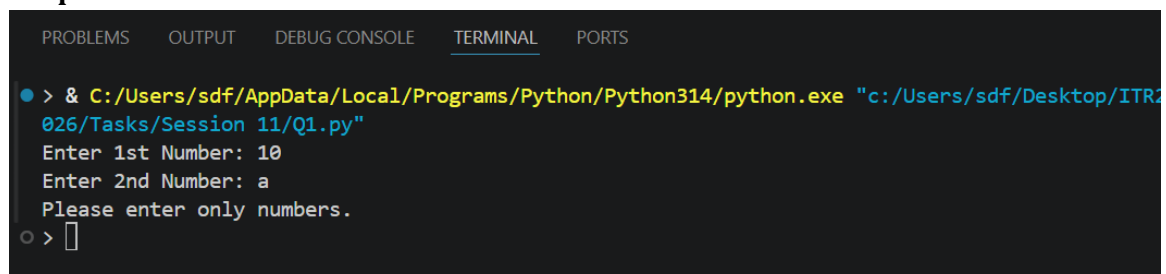
Program Code:

```
try:
    a = int(input("Enter 1st Number: "))
    b = int(input("Enter 2nd Number: "))

    c = a / b
    print("Division:", c)

except ValueError:
    print("Please enter only numbers.")
```

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2
026/Tasks/Session 11/Q1.py"
Enter 1st Number: 10
Enter 2nd Number: a
Please enter only numbers.
> 
```

Q2

Q2. Handling ZeroDivisionError)

Write a program that divides two numbers entered by the user.

Use try-except to specifically catch ZeroDivisionError and print:

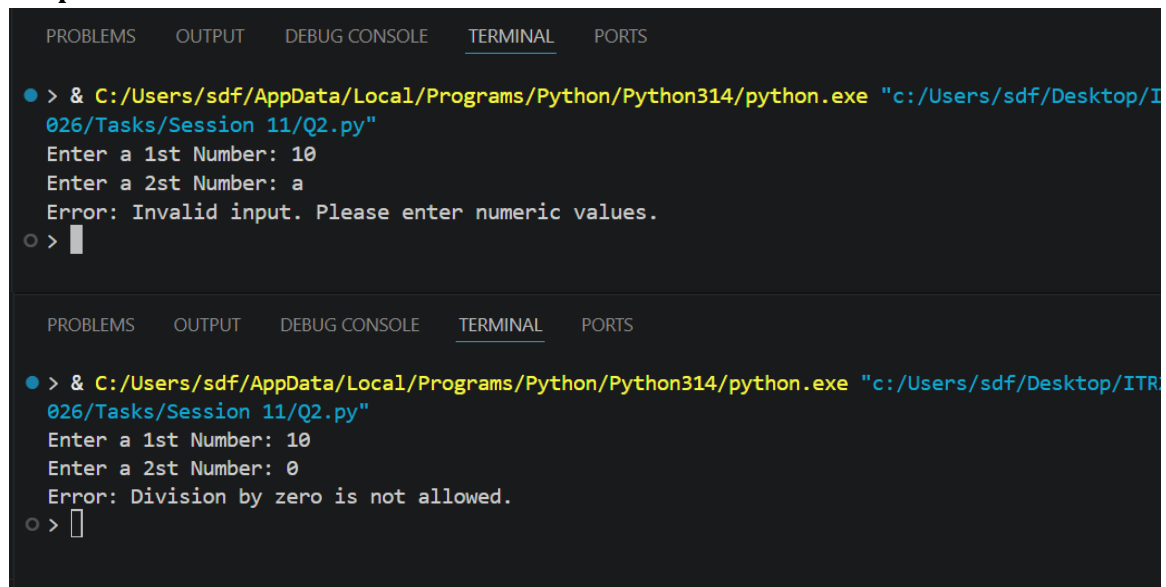
"Error: Division by zero is not allowed."

Also handle ValueError for invalid input.

Program Code:

```
try:
    a = int(input("Enter a 1st Number: "))
    b = int(input("Enter a 2st Number: "))
    c = a / b
    print("Division :", c)
except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")
except ValueError:
    print("Error: Invalid input. Please enter numeric values.")
```

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR
026/Tasks/Session 11/Q2.py"
Enter a 1st Number: 10
Enter a 2st Number: a
Error: Invalid input. Please enter numeric values.
○ > █

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR
026/Tasks/Session 11/Q2.py"
Enter a 1st Number: 10
Enter a 2st Number: 0
Error: Division by zero is not allowed.
○ > █
```

Q3

Q3. Multiple except Blocks)

Create a program that performs division and also tries to convert a string to integer.

Handle both ValueError and ZeroDivisionError with separate except blocks and appropriate messages.

Program Code:

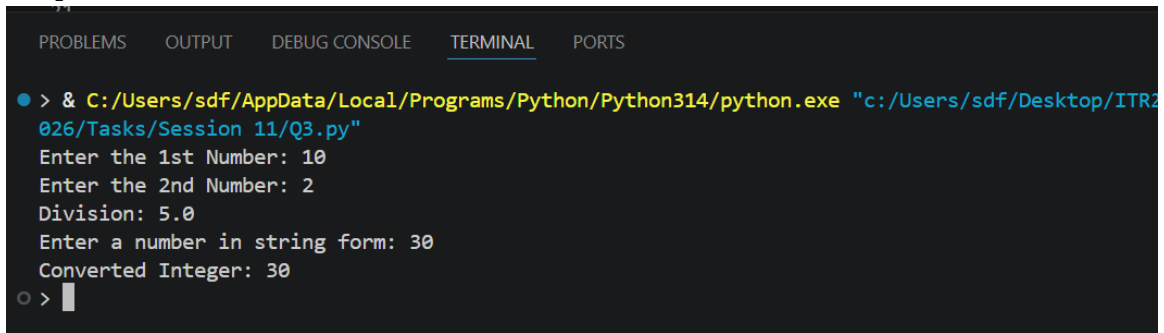
```
try:
    # Division
    a = int(input("Enter the 1st Number: "))
    b = int(input("Enter the 2nd Number: "))
    print("Division:", a / b)

    # String to Integer Conversion
    s = input("Enter a number in string form: ")
    num = int(s)
    print("Converted Integer:", num)

except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")

except ValueError:
    print("Error: Invalid input. Please enter a valid number.")
```

Output Screenshot:



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there is a navigation bar with five tabs: "PROBLEMS", "OUTPUT", "DEBUG CONSOLE", "TERMINAL" (which is selected and underlined), and "PORTS". Below the tabs, the terminal displays the following text:

```
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 11/Q3.py"
Enter the 1st Number: 10
Enter the 2nd Number: 2
Division: 5.0
Enter a number in string form: 30
Converted Integer: 30
○ > █
```

The prompt "● >" is followed by the command to run the Python script. The script prompts the user for two numbers, performs a division, and then prompts for a string number which is converted to an integer. The terminal shows the user's input and the program's output. At the bottom, there is a new prompt "○ >" with a cursor.

Q4

Q4. Using else)

Write a program that takes two numbers and divides them.

Use try, except, and else.

The else block should print "Division performed successfully!" only when no exception occurs.

Program Code:

try:

```
a = int(input("Enter a 1st Number: "))
```

```
b = int(input("Enter a 2st Number: "))
```

```
c = a / b
```

```
print("Division :", c)
```

except ZeroDivisionError:

```
print("Error: Division by zero is not allowed.")
```

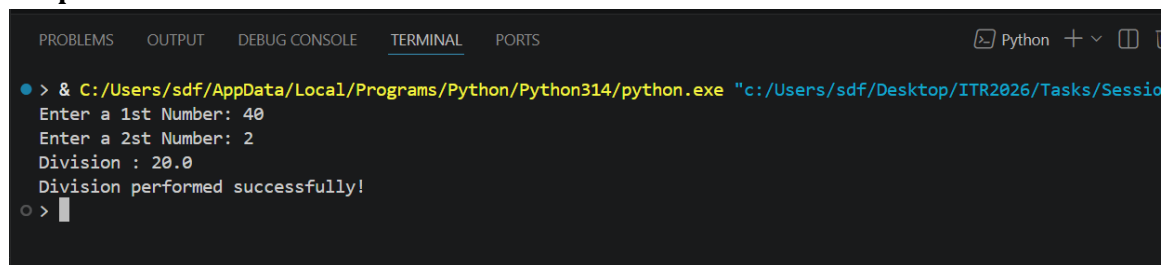
except ValueError:

```
print("Error: Invalid input. Please enter numeric values.")
```

else:

```
print("Division performed successfully!")
```

Output Screenshot:



The screenshot shows a Python IDE interface with a terminal window. The terminal displays the following output:

```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 4/Python/Python314/python.exe"
Enter a 1st Number: 40
Enter a 2st Number: 2
Division : 20.0
Division performed successfully!
>
```

Q5

Q5. Using finally)

Modify the division program to include a finally block that always prints:

"Program execution completed."

Demonstrate that finally runs whether an exception occurs or not.

Add comments explaining the purpose of finally.

Session 11 (AIML) - Assignment Questions 1

Program Code:

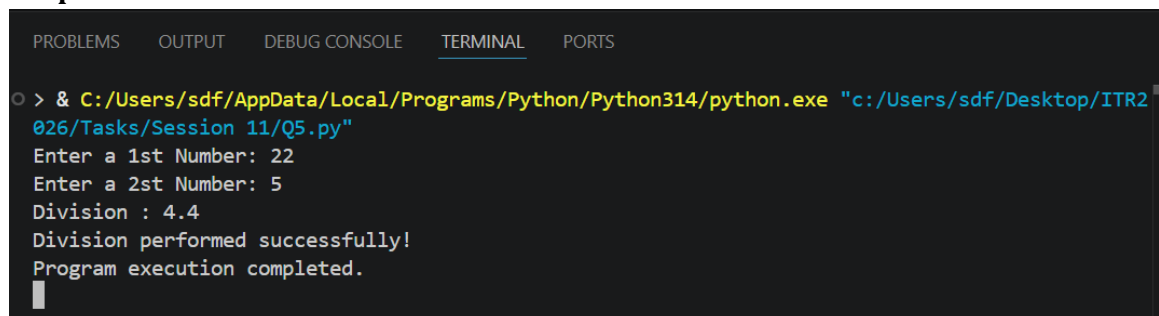
```
try:
    a = int(input("Enter a 1st Number: "))
    b = int(input("Enter a 2st Number: "))
    c = a / b
    print("Division :", c)

except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")
except ValueError:
    print("Error: Invalid input. Please enter numeric values.")

else:
    print("Division performed successfully!")

finally:
    print("Program execution completed.")
```

Output Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2
026/Tasks/Session 11/Q5.py"
Enter a 1st Number: 22
Enter a 2st Number: 5
Division : 4.4
Division performed successfully!
Program execution completed.
```

Q6

Q6. Combined try-except-else-finally)

Write a complete division program using all four blocks (try, except, else, finally):

Handle ValueError and ZeroDivisionError.

Print success message in else.

Always thank the user in finally.

Program Code:

```
try:
    a = int(input("Enter a 1st Number: "))
    b = int(input("Enter a 2st Number: "))
    c = a / b
    print("Division :", c)

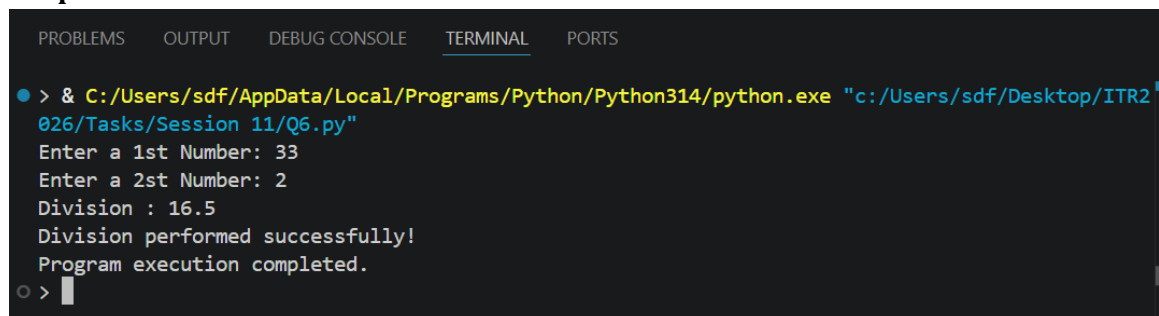
# Handle ValueError and ZeroDivisionError.
except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")

except ValueError:
    print("Error: Invalid input. Please enter numeric values.")

else:
    print("Division performed successfully!")

finally:
    print("Program execution completed.")
```

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2
026/Tasks/Session 11/Q6.py"
Enter a 1st Number: 33
Enter a 2st Number: 2
Division : 16.5
Division performed successfully!
Program execution completed.
>
```


Q7

Q7. Practical - Age Input)

Write a program that asks the user for their age.

Convert the input to integer.

If the age is negative, raise a ValueError with message "Age cannot be negative."

Handle the exception gracefully and print a suitable message.

Program Code:

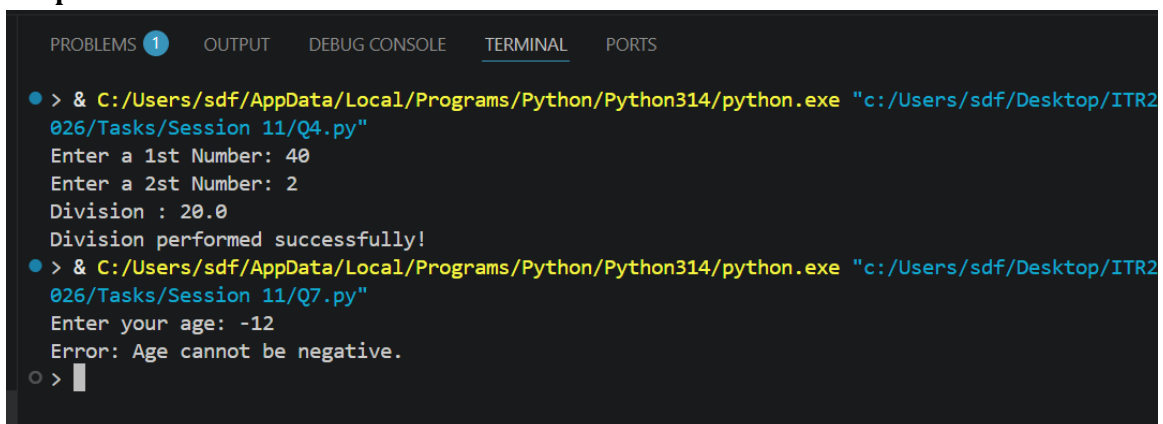
```
try:
    age = int(input("Enter your age: "))

    if age < 0:
        raise ValueError("Age cannot be negative.")

    print("Your age is:", age)

except ValueError as e:
    print("Error:", e)
```

Output Screenshot:



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

• > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 11/Q4.py"
Enter a 1st Number: 40
Enter a 2st Number: 2
Division : 20.0
Division performed successfully!

• > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 11/Q7.py"
Enter your age: -12
Error: Age cannot be negative.

○ > █
```

Q8

Q8. Multiple Exceptions in One Block)

Create a program that:

Takes a number as input.

Divides 100 by that number.

Handle both ValueError and ZeroDivisionError in a single except block using a tuple.

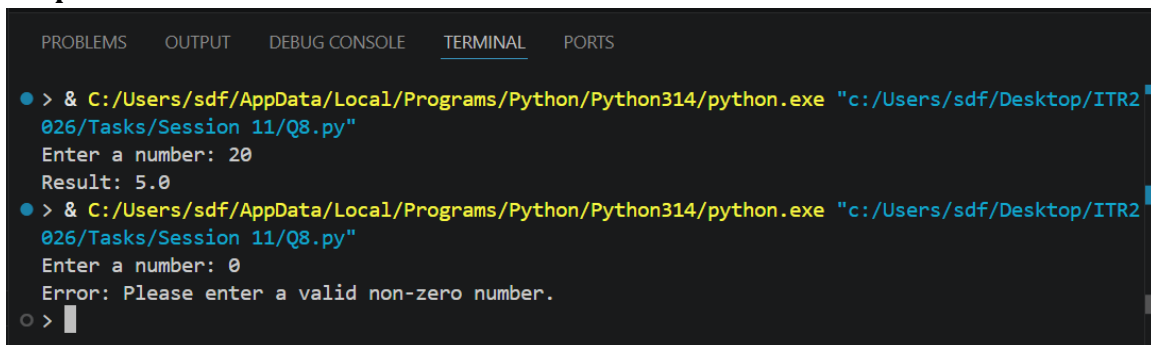
Print a combined error message.

Program Code:

```
try:
    num = int(input("Enter a number: "))
    result = 100 / num
    print("Result:", result)

except (ValueError, ZeroDivisionError):
    print("Error: Please enter a valid non-zero number.")
```

Output Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2
026/Tasks/Session 11/Q8.py"
Enter a number: 20
Result: 5.0
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2
026/Tasks/Session 11/Q8.py"
Enter a number: 0
Error: Please enter a valid non-zero number.
○ > █
```

Q9

Q9. Debugging Exception Code)

The following code has issues. Correct it and add comments explaining the fixes:

Python

```
try
num = int(input("Enter a number: "))
result = 100 / num
print("Result:", result)
except ValueError
print("Please enter a valid number")
Session 11 (AIML) - Assignment Questions 2
except:
print("Some error occurred")
```

Program Code:

```
try:
    # Take input from the user and convert it to an integer
    num = int(input("Enter a number: "))

    # Divide 100 by the entered number
    result = 100 / num

    # Display the result
    print("Result:", result)

# Fix 1: Added a colon(:) after except ValueError
except ValueError:
    # This block runs if the user enters a non-numeric value
    print("Please enter a valid number.")

# Fix 2: Used a general except block with a colon (:)
except:
    # This block handles any other unexpected errors
    print("Some error occurred.")
```

Output Screenshot:



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), and 'PORTS'. In the top right corner, it says 'Python: Q9' with a plus sign and a square icon. The terminal content shows a command prompt with a blue dot icon, followed by a command to run a Python script. The script prompts for a number, the user enters '2', and the script outputs 'Result: 50.0'. A second prompt is visible at the bottom.

```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 9/Task 9.1/Task 9.1.py"
Enter a number: 2
Result: 50.0
> 
```

Q10

Q10. Mini Project - Exception Handling)

Create a Simple Calculator with exception handling:

The program should run in a loop and show this menu:

text

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Take user choice and two numbers.

Perform the operation inside try block.

Handle ValueError (invalid input) and ZeroDivisionError (for division).

Use finally to print "Operation attempted." after every calculation.

Continue until user chooses to exit.

Session 11 (AIML) - Assignment Questions 3

Program Code:

```
while True:
    print("\n===== Simple Calculator =====")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. Exit")

    choice = input("Enter your choice (1-5): ")

    if choice == "5":
        print("Thank you! Calculator Closed.")
        break

    try:
        num1 = int(input("Enter First Number: "))
        num2 = int(input("Enter Second Number: "))

        if choice == "1":
            print("Result:", num1 + num2)

        elif choice == "2":
```

```
    print("Result:", num1 - num2)

elif choice == "3":
    print("Result:", num1 * num2)

elif choice == "4":
    print("Result:", num1 / num2)

else:
    print("Invalid Choice! Please enter 1 to 5.")

except ValueError:
    print("Error: Please enter valid numbers.")

except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")

finally:
    print("Operation attempted.")
```

Output Screenshot:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR
026/Tasks/Session 11/Q10.py"

===== Simple Calculator =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 1
Enter First Number: 12
Enter Second Number: 6
Result: 18
Operation attempted.

===== Simple Calculator =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 2
Enter First Number: 10
Enter Second Number: 3
Result: 7
Operation attempted.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR/026/Tasks/Session 11/Q10.py"
```

```
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 2
Enter First Number: 10
Enter Second Number: 3
Result: 7
Operation attempted.
```

```
===== Simple Calculator =====
```

```
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 3
Enter First Number: 10
Enter Second Number: 3
Result: 30
Operation attempted.
```

```
===== Simple Calculator =====
```

```
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 5
Thank you! Calculator Closed.
```

o >